

Ex 110

$$X \perp\!\!\!\perp Y, X \sim Y, X(\Omega) \subset \mathbb{N}, Y(\Omega) \subset \mathbb{N}$$

$$Z \sim \mathcal{G}(p), Z = X + Y + 1$$

1) Par linéarité de l'espérance,

$$E(Z) = E(X) + E(Y) + 1 = 1 + 2E(X) \text{ car } X \sim Y$$

$$\text{Donc } E(X) = \frac{E(Z) - 1}{2} = \boxed{\frac{1 - p}{2p} = E(X)}$$

$$X \perp\!\!\!\perp Y \text{ donc } V(X + Y + 1) = V(X) + V(Y) = 2V(X) = V(Z)$$

$$\text{Donc } V(X) = \frac{V(Z)}{2} = \boxed{\frac{1 - p}{2p^2} = V(X)}$$

2) $Z \sim \mathcal{G}(p)$ donc $\forall t \in \left] -\frac{1}{1-p}, \frac{1}{1-p} \right[$,

$$G_Z(t) = \frac{p e^t}{1 - (1-p)e^t} = E(t^Z) = E(t^{X+Y+1}) = t E(t^{X+Y}) = t G_{X+Y}(t)$$

$$X \perp\!\!\!\perp Y \text{ donc } G_{X+Y} = G_X G_Y = G_X^2 \text{ car } X \sim Y.$$

$$\text{Donc } \forall t \in \left] -\frac{1}{1-p}, \frac{1}{1-p} \right[$$

$$G_X(t) = \sqrt{\frac{p}{1 - (1-p)t}}$$

On prend $G_X > 0$
car $G_X(1) = 1$

3) $\forall t \in \left] -\frac{1}{1-p}, \frac{1}{1-p} \right[$,

$$(1 - (1-p)t)^{-1/2} = \sum_{n=1}^{+\infty} \left(\frac{1}{n!} \left(-\frac{1}{2}\right) \dots \left(-\frac{1}{2} - i\right) \frac{(-(1-p)t)^n}{n!} \right) + 1$$

$$\begin{aligned}
 (1 - (1-p)t)^{-1/2} &= 1 + \sum_{n=1}^{+\infty} \frac{1}{2} \times \frac{3}{2} \times \dots \times \frac{2n-1}{2} \frac{(1-p)^n}{n!} t^n \\
 &= \sum_{n=0}^{+\infty} \frac{(2n)!}{4^n (n!)^2} (1-p)^n t^n
 \end{aligned}$$

Donc $G_X(t) = \sqrt{p} \sum_{n=0}^{+\infty} \frac{(2n)!}{4^n (n!)^2} (1-p)^n t^n$

Donc $\forall n \in \mathbb{N}, P(X=n) = \frac{(2n)!}{4^n (n!)^2} \sqrt{p} (1-p)^n$